

ELTON SAKYI

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PROFESSIONAL SUMMARY

Computer Science graduate from Winthrop University and software developer focused on backend systems, full-stack web applications, database-driven platforms, and API-aware product architecture. Hands-on experience building Laravel/PHP, React, TypeScript, Node, Java, SQL, and AWS-based projects, including multi-role systems for healthcare-style camp operations, structured public-data platforms, and personal finance decision support. Strong interest in secure software design, API integration, data modelling, networking, and production-quality engineering practices.

EDUCATION

Winthrop University *B.S. Computer Science | Minor in Chemistry*

Relevant Coursework: Algorithm Analysis & Data Structures, Software Engineering, Advanced Programming, Computer Architecture, Operating Systems, Computer Science Theory, Database Processing, Cloud Computing, Calculus I & II, Discrete Mathematics, Network Processing, Physics with Calculus, Analytical Chemistry, Inorganic Chemistry, Chemometrics.

University of South Carolina Lancaster *Associate in Science*

SOFTWARE ENGINEERING EXPERIENCE

Camp Burnt Gin Application System *Full-Stack Developer / Systems Engineer*

2026

Laravel, React, TypeScript, MySQL, AWS, Vercel, GitHub

- Developed and improved a full-stack camp application and management platform with Applicant, Admin, Super Admin, and Medical Provider workflows.
- Built and refined modules for digital applications, paper application intake, camper records, medical records, document uploads and review, inbox messaging, notifications, reporting, dashboards, audit logging, role-based permissions, and session management.
- Worked across Laravel 12 backend services, React + TypeScript frontend pages, database migrations, seeders, policies, controllers, API resources, and validation logic.
- Improved reliability by auditing frontend/backend mismatches and fixing draft vs. submitted application behaviour, document compliance recognition, camper deletion edge cases, paper-form workflows, and medical portal data flows.
- Supported deployment workflows using AWS EC2, Nginx, PHP-FPM, MySQL, Node/Vite builds, Vercel previews, GitHub version control, and environment-specific configuration.

Pathfinder-X *Founder / Lead Developer*

2025

Laravel, MySQL, REST APIs, Postman, GitHub, Public Data Integrations

- Designed and built a modular full-stack platform that organizes structured academic, occupational, and labor-market information through backend services and external public API integrations.
- Implemented a normalized database schema, service-layer logic, validated REST API endpoints, and clean integration of external public data sources.
- Engineered modular backend services with clean domain boundaries to support scalable data retrieval and long-term maintainability.
- Followed disciplined engineering practices including Postman API testing, normalized data modeling, and structured internal documentation.

Nexus *Backend Architect / Lead Developer*

2026

Property Rental & Financial Workflow Platform | Laravel, MySQL, REST APIs, RBAC, Stripe-style payments

- Architected a Ghana-focused property rental platform covering listings, tenant onboarding, lease lifecycle management (application → active lease → renewal/end), and ledger-style financial tracking for rent charges, payments, balances, and fees.
- Designed a multi-role RBAC system with distinct permissions for landlords, tenants, property managers, and admins, ensuring users only access and act on what their role allows.
- Built modular backend services with clean domain boundaries: property, tenant, lease, payment, and reporting logic each isolated to support maintainability and reliability at scale.
- Engineered an analytics-ready database schema supporting rent collection trends, tenant activity, property performance, outstanding balances, and payment history.
- Designed payment flows around Stripe-style concepts with traceable, auditable transaction records to maintain financial integrity across the platform.

Hackathon: FutureYou *Personal Finance Decision Engine*

2026

React/Vite, Laravel/PHP, AI-assisted workflows

- Built and presented a forward-looking personal finance concept centred on the loop: decision, consequence, action.
- Designed a frontend-first experience that lets users test financial choices and see projected effects on savings, debt, goals, and long-term outcomes.
- Outlined a dual-layer insight model combining rule-based calculations for instant feedback with AI-generated explanations for clearer financial literacy.

Time Loop Snake *Full-Stack Developer*

2026

MERN Platform with Canvas Game | React 18, Node 20, Express 4, MongoDB Atlas, AWS EC2, Nginx, PM2

- Built a MERN-stack web application around a Canvas-based Snake game featuring a time-loop mechanic: every 14 seconds, a ghost of the player's past self spawns and replays their exact moves.
- Implemented an Express REST API with JWT authentication, bcrypt password hashing (12 rounds), role-based access control, and ownership checks; layered accounts, authenticated score submission, a public leaderboard, and an admin user/score management panel on top of the game.
- Modeled data in MongoDB Atlas via Mongoose 8 with validated schemas and indexes; served a React 18 SPA via Vite and React Router 6.
- Hardened the API with helmet, CSP, HSTS, CORS allowlisting, rate limiting, mongo-sanitize, and a 10kb body limit.
- Deployed to AWS EC2 (Ubuntu) behind an Nginx reverse proxy with HTTPS, PM2 process supervision, and a GitHub Actions CI/CD pipeline triggered on push to main.

SELECTED PROGRAMMING PROJECTS

Partial Compiler for a Pascal-like Language *Systems Programming & Language Processing*

Lex, YACC, C, Regular Expressions, Context-Free Grammars

- Developed a partial compiler that translates a simplified Pascal-like high-level language into pseudo-assembly code.
- Created a lexical analyzer with Lex to tokenize keywords, variables, numbers, operators, assignment symbols, and invalid characters.
- Built grammar rules with YACC and context-free grammars to parse assignments, arithmetic expressions, while loops, and if-then / if-then-else conditional statements.
- Added syntax validation with line-number error reporting to surface invalid program structures during compilation.

Movie Theatre Ticket Sales System *PHP Full-Stack Web Application*

PHP, MySQL, HTML, CSS, Apache/XAMPP, GitHub

- Created a secure ticketing application with user login, movie browsing, seat purchasing, admin CRUD features, MySQL records and logs, password hashing, and SQL injection prevention.

Secure File Sharing Platform *PHP Web Application*

PHP, HTML, CSS, XAMPP, GitHub

- Built a secure file exchange tool with file validation, session-based access control, directory traversal protection, and local deployment through XAMPP.

AWS & SQL Practical Work *Cloud Infrastructure & Database Engineering*

AWS Management Console, AWS CLI, RDS, MySQL Workbench, SQLTools

- Worked with cloud and database concepts including SQL joins, grouping, subqueries, indexing, RDS configuration, IAM permissions, MFA, and role-based access practices.

TECHNICAL SKILLS

Programming Languages: Java, PHP, C++, C#, JavaScript/TypeScript, SQL, MIPS Assembly, HTML, CSS

Frameworks & Platforms: Laravel, React, Vite, Node/Express, XAMPP/Apache, MySQL, MongoDB

Backend & APIs: REST API design, authentication, authorization/RBAC, validation, service layers, database migrations, seeders, API testing

Cloud & Deployment: AWS EC2, AWS RDS, IAM, Nginx, PHP-FPM, Vercel, environment configuration, production/pre-production workflows

Tools: Git, GitHub, Postman, VS Code, Eclipse, IntelliJ, Visual Studio, MySQL Workbench, SQLTools, Composer, pnpm/npm, Wireshark

Academic & Analysis: Mathematica, LaTeX, Microsoft Office, Excel, Google Sheets

RESEARCH & WRITING

Digital Dependency: Cognitive and Neurological Consequences of Technology Use *CSCI 327 Research Paper | Winthrop University 02/2025*

- Analyzed cognitive, neurological, and societal effects of prolonged digital technology use, including attention, memory, and brain-health concerns.

Ethical Responsibilities of Technology Corporations in the Digital Age *CSCI 327 Research Paper | Winthrop University 04/2025*

- Examined persuasive design, data collection, addictive features, user autonomy, digital well-being, and governance responsibilities for technology corporations.

ADDITIONAL INFORMATION

Languages: Native bilingual in English and Akan; working proficiency in German.

Memberships: USC Lancaster Chemistry Club; G.E.N.T.S.; Winthrop University Chemical Society; Winthrop Undergraduate Research Organization; Winthrop University Chess Club.